Topical 5-Fluorouracil and Imiquimod for the Treatment of High-Grade Cervical Intraepithelial Neoplasia: Protocol for a Randomised Open-Label Controlled Trial (TOPFIT-CIN Study)

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Abstract

Background

There has been a long-standing interest in developing a safe, effective topical therapy for treating high-grade cervical intraepithelial neoplasia (CIN2/3). Combining 5-Fluorouracil (5-FU) and Imiquimod as a topical application may synergistically target CIN lesions while minimising adverse effects associated with excisional procedures. However, more research is still needed to understand the potential of these agents when combined as a treatment for CIN 2/3.

Aim

The *"TOPFIT-CIN"* study will assess the efficacy and safety of topical 5-FU/imiquimod compared to loop electrosurgical excision procedure (LEEP), as standard of care, in treating patients with CIN 2/3.

Methods

This is a study protocol for a non-inferiority, randomised, open-label, controlled trial that will be conducted at the gynecologic oncology, cytology and colposcopy clinics of the Lagos University Teaching Hospital (LUTH) in Lagos, Nigeria over two years (July 2025 to June 2027). At baseline, n = 90 sexually active women aged 30–65 years and diagnosed with high-grade CIN lesions (CIN 2 or CIN 3) confirmed by colposcopy biopsy will be randomised to receive either weekly topical application of 5-FU/Imiquimod cream or LEEP. The primary endpoint is the proportion of participants achieving treatment response (complete regression of CIN2/3 lesions at 3 months post-treatment initiation) between treatment arms. The data analysis will be conducted on an intention-to-treat and per-protocol basis. The comparison of treatment response and secondary outcomes (abnormal cervical cytology reversal, HPV clearance, adverse events, and CIN2/3 recurrence) between the two treatment arms will be performed using the generalised linear model to estimate the relative risks (RR) and 95% confidence intervals (95% CI). Statistical significance will be reported as $P < 0.05$.

Discussion

The "*TOPFIT-CIN"* study will evaluate the efficacy of topical 5-FU/ imiquimod compared to LEEP in treating patients with CIN 2/3. If proven effective, topical 5-FU and imiquimod combination therapy could offer a non-invasive alternative to traditional excisional procedures, particularly in resource-limited settings where access to specialised surgical care is limited. In addition, by mitigating the risks associated with invasive procedures such as LEEP and cold-knife conisation (CKC), this new treatment approach could improve patient outcomes, preserve fertility, and alleviate the economic burden of cervical cancer treatment on healthcare systems worldwide.

Registration: PACTR202505720883384 (12th May 2025).

Introduction

Invasive cervical cancer (ICC) remains a leading cause of cancer-related morbidity and mortality among women worldwide, particularly in low- and middle-income countries where access to screening and treatment is limited [1]. High-grade cervical intraepithelial neoplasia (CIN2/3) is a premalignant condition of the cervix caused by infection with high-risk human papillomavirus (HPV) [2]. CIN2/3 lesions, if left untreated, can progress to ICC [3]. While excisional procedures such as loop electrosurgical excision procedure (LEEP) and cold knife conisation (CKC) are effective in treating CIN2/3 [3], they are associated with risks of complications including bleeding, infection, and cervical stenosis. They can lead to adverse obstetric outcomes, including cervical insufficiency and preterm labour, in future pregnancies [4]. They are also associated with significant individual and infrastructure costs [5]. A non-surgical or medical treatment option with the potential for patient self-application could overcome many barriers to surgical therapy, including desire for childbearing, cost, health provider skill, equipment, location, and patient fear [6, 7]. Additionally, primary medical treatment would address a treatment bottleneck in resource-limited settings by allowing immediate or point-of-care treatment of CIN2/3 [8].

There has been a long-standing and continued interest in the development of a safe, effective topical therapy for treating CIN2/3 and HPV [8]. Recent studies have explored the effectiveness of topical agents such as 5-fluorouracil (5-FU) and imiquimod for treating CIN2/3 [9–11]. 5-FU, a pyrimidine analogue and antiproliferative agent, inhibits DNA synthesis and has been shown to induce apoptosis in cervical neoplastic cells [12]. Modified regimens have shown better patient tolerance in treating recurrent and primary cervical dysplasia [13, 14]. Imiquimod, a toll-like receptor agonist, stimulates local immune responses and increases the production of cytokines interferon α (IF-α), interleukin 6 (IL-6), and tumour necrosis factor α (TNF-α). It has demonstrated efficacy in treating genital warts caused by human papillomavirus (HPV) [15]. Combining these agents (5-FU and Imiquimod) may offer a synergistic effect in targeting CIN lesions while minimising adverse effects associated with excisional procedures. However, more research is still needed to understand the potential of these agents when combined as a treatment for CIN 2/3.

The central hypothesis of our proposed study is that the combined use of 5-FU and imiquimod, two widely available, low-cost generic drugs, can be repurposed as an alternative non-surgical primary treatment approach for CIN2/3 to reduce the risk of persistent/recurrent CIN2/3 and progression to ICC. The *TOPFIT-CIN* study, therefore, will assess the efficacy of topical 5-FU/ imiquimod compared to LEEP (as standard of care) in treating patients with CIN 2/3.

Patients and Methods

Study design and setting

The "*TOPFIT-CIN study*" is a non-inferiority, open-label, randomised controlled trial that will be conducted at the gynecologic oncology, cytology and colposcopy clinics of the Lagos University Teaching Hospital (LUTH) in Lagos, Nigeria over two years (July 2025 to June 2027). LUTH is the leading healthcare institution in Lagos, acting as a referral centre for other public and private hospitals in Lagos and nearby states of Ogun and Oyo. The hospital offers gynecologic oncology services including cervical cancer screening and treatment care such as Pap smears, human papillomavirus (HPV) testing, colposcopy, and pathological services, including cytology and histology, and standard treatment with cryotherapy, thermal ablation therapy (TAT), cold knife conization (CKC) or loop electrosurgical excision procedure (LEEP) [16]. The main study clinic (colposcopy clinic) operates every weekday and provides cervical cancer screening and treatment services to ~ 250 women each month.

Study population

Potentially eligible participants in the *TOPFIT-CIN* study are sexually active women aged 30–65 years and diagnosed with high-grade CIN lesions (CIN 2 or CIN 3) confirmed by colposcopic biopsy after positive cervical cytology and/or high-risk HPV testing. Inclusion criteria include women free from any mental or physical disabilities that inhibit them from understanding the implications of the study and those not considering relocating from their current residence within the next year. Exclusion criteria are suspicious cervical lesions, concomitant vulvar and/or vaginal intraepithelial neoplasia, immunodeficiency, current pregnancy or within 6 weeks of childbirth, previous hysterectomy, or history of CC or therapy for benign or malignant cervical lesions, current malignant disease, refusal of consent at enrolment, and inability to self-collect adverse event data using the symptom diary or a visual analogue scale (VAS). Women who are confirmed as having CC at any time during this study will be referred for treatment and excluded from further participation.

Study endpoints and sample size calculation

The study’s primary endpoint is the proportion of participants achieving complete regression of CIN2/3 lesions, defined as negative colposcopy and biopsy findings, at 3 months post-treatment initiation between treatment arms. The secondary endpoints include the rates of reversal of cervical cytological abnormalities, high-risk human papillomavirus (HR-HPV) clearance, adverse event (safety) profiles, and CIN2/3 recurrence between the two treatment arms. Assuming a randomisation ratio of 1:1, a one-sided test with a type I error of 0.05, and a 15% loss to follow-up or missing outcome data rate, we planned to enrol 90 participants with CIN2/3. This will power the trial at 80% to detect a non-inferiority margin of 12% for CIN2/3 regression, given a treatment response rate of 85% in the 5-FU/imiquimod arm and 95% in the LEEP arm [17]. Given a CIN2/3 global prevalence of 4.3% [18] and a 5% non-response rate, a total of n = 2,204 women will be required for screening with Pap smear/HR-HPV co-testing to achieve the sample size of n = 90 patients with CIN2/3 sufficient to assess the efficacy of topical 5-FU/ imiquimod in treating CIN2/3.

Participants screening

To achieve the maximum sample size required, the study team will screen and enrol n = 2,204 consecutively consenting women attending routine care at the adult general outpatient clinics of the hospital. Study intent and procedures will be introduced, and informed consent will be obtained before any study procedure. Once consent is obtained, baseline sociodemographic, sexual and clinical information will be collected using an electronic case report form on REDCap. The women will then undergo cervical liquid-based cytology (LBC) sample collection using a Cytobrush that is broken off and placed into a ThinPrep vial containing PreservCyt Solution (Hologic Inc., Marlborough, MA 01752, USA) for human papillomavirus (HPV) DNA and Pap smear cytology co-testing. HPV DVA testing will be performed using Polymerase Chain Reaction (PCR) that targets HPV DNA, followed by nucleic acid hybridisation to detect 14 high-risk HPV genotypes – HPV types 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 66 and 68 [2]. After the HPV DNA testing, the PreservCyt residual LBC material will be stored in the original ThinPrep vials at room temperature and analysed for Pap smear cytology. A Pap smear cytology slide will be prepared and analysed. The result will then be classified according to the revised Bethesda's guideline as negative for intraepithelial lesion and malignancy (NILM), atypical squamous cells of undetermined significance (ASCUS), atypical squamous cells cannot exclude HSIL (ASC-H), low-grade squamous intraepithelial lesion (LSIL) and high-grade squamous intraepithelial lesion (HSIL) [19]. All eligible participants with HPV DNA-positive or abnormal Pap smear cytology tests are recalled for colposcopic evaluation and cervical biopsy. Lesions suspicious for CIN, as per usual care, are biopsied, and in the absence of that, a four-quadrant biopsy will be performed, and the samples are then collected in a 10% buffered formalin-containing recipient for transportation to the Anatomic and Molecular Pathology (AMPATH) laboratory for histologic analysis. A trained Pathologist will review the prepared histology slides, and the results will then be classified as normal, low-grade (CIN1), or high-grade (CIN2/3) lesions, or invasive cervical cancer (ICC). Women with histologically confirmed CIN2/3 lesions will be randomised into the study treatment arms.

Randomisation and allocation concealment

Using a computer-generated random sequence by the study statistician, participants with CIN2/3 will be randomly assigned by the investigators in a ratio of 1:1 to either: (1) alternate weekly topical application of 5-FU cream on weeks 1, 3, 5 and 7 and Imiquimod cream on weeks 2, 4, 6 and 8 or (2) LEEP as the standard of care [Figure 1]. Allocations to treatment arms will be concealed from the investigators and trial statistician using sealed opaque envelopes that are sequentially numbered. The envelopes will be prepared in advance of the trial and will only be opened sequentially at the time of participants' enrollment, ensuring that the treatment allocation remains concealed until the point of assignment.

Treatment procedures

- *Topical 5-FU/Imiquimod arm*: A sterile vaginal speculum will be inserted to visualize the participant's cervix clearly, after which the topical cream is administered by a trained study nurse or physician from the day of randomization by use of a vaginal applicator in a dose of 2g of 5% cream for 5-FU (on Day 1, 15, 29 and 43) and 12.5mg in 250mg cream for Imiquimod (on Day 8, 22, 36, and 50). The remainder of the cream will be rinsed in the shower the next morning. The use of tampons or other sanitary materials is only allowed from that moment to prevent local side effects. In the topical cream treatment arm, the cream application is delayed until the end of menses if a participant reports menses that coincide with an application date. There should always be a minimum of seven days' interval between each treatment application. Participants will also be asked to avoid sexual activity for 48 hours after application of the study medication.

- *LEEP arm*: After inserting a sterile vaginal speculum is inserted to visualise the participant's cervix, a colposcope is used to identify the transformation zone (TZ) and the abnormal areas of the cervix requiring excision. An injection of 1% lidocaine with or without adrenaline (to reduce bleeding) is administered around the cervix, particularly at the 3, 6, 9, and 12 o'clock positions. After a few minutes, the cervical excision is performed by attaching an appropriate-sized wire loop electrode to the electrosurgical unit and then the diathermy settings (typically 30–50 watts for cutting mode) are adjusted for excision of the transformation zone and visible lesion in one smooth pass using the loop electrode. The excised tissue should include the entire transformation zone and the abnormal areas (to allow histopathological evaluation). Electrosurgical coagulation using a ball electrode is applied to bleeding areas (using lower wattage) to address any bleeding points. Additional haemostasis will be achieved using Monsel's solution (ferric subsulfate) or silver nitrate sticks. The participant is monitored for about 4 hours for any immediate adverse effects or complications and is then asked upon discharge to avoid sexual intercourse, tampon use, and douching for 4 weeks to allow proper healing.

To ensure a collaborative approach that can make the spouse feel involved and supportive in the treatment process, the following measures will be adopted: (1.) Scheduling an orientation session or counselling at the beginning of the treatment to educate both the participant and their spouse about the treatment procedures, rationale, and expectations and (2.) Encouraging the spouse to help track treatment days (randomisation days, application schedule, and menstrual adjustments) during the treatment period.

Follow-up and adverse events evaluation

Participants with abnormal Pap smear cytology or who test positive for any HR-HPV DNA types at 3- or 6-month post-treatment follow-ups will undergo colposcopy evaluation with lesion-targeted or four-quadrant biopsy (as indicated) [Table 1]. Women who do not respond adequately to topical treatments or have CIN lesions at 3 months will be offered surgical treatment with LEEP at no cost and are then excluded from further participation in the study. At baseline, participants will be trained on the recognition and recording of adverse events and are then asked to document the occurrence and severity of genital/local (including blisters, ulcerations, pustules or pruritus lasting more than 5 days) or systemic adverse events (flu-like symptoms, fatigue or any unusual symptoms) daily during treatment using a symptom diary and a visual analogue scale (VAS). Study visits will be scheduled at weeks 2, 4, 6, and 8, during which safety (adverse events) are assessed and at 4–6 weeks after the last topical application (to coincide with the 3rd month) and 26 weeks (6th month) to assess treatment response

(CIN2/3 regression) and CIN2/3 recurrence respectively using cervical Pap smear cytology and HR-HPV co-testing. Safety outcomes will also be defined and assessed using the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) version 4.0 for non-genital events and the Female Genital Grading Table (FGGT) [20] as appropriate. In women with grade 1 adverse events, treatment will be continued as long as the woman is willing to continue study participation. In patients with grade 2 and persistent systemic or local adverse events, treatment will be suspended for 7 days, and then a reassessment is performed to determine if the next topical application could be restarted or completely discontinued. In patients with grade 3 or 4 adverse events, treatment would be suspended and a LEEP scheduled as soon as possible. In case of systemic drug-related adverse events, participants will be prescribed paracetamol and nonsteroidal anti-inflammatory drugs. All AEs, regardless of severity or suspected relationship to the intervention, will be documented and reviewed by the study safety team and DSMC. Serious adverse events (SAEs) will be reported to the ethics committee per institutional and national guidelines. For women who miss their follow-up visits, the study team will make multiple attempts to contact them via their preferred communication methods (e.g., phone calls, text messages, or emails) to remind them of the missed visit and reschedule it at their earliest convenience. If the participant cannot attend the clinic for their non-treatment visit due to logistical or medical reasons, the study team will arrange a home visit (if feasible) or telehealth consultation to collect essential safety data and will be asked to submit their completed symptom diary and VAS remotely (e.g., via WhatsApp or email). If there are concerns about adverse events, participants will be encouraged to report any symptoms experienced during the missed visit period. Participants who fail to attend more than two treatment visits or 50% of non-treatment follow-up visits despite efforts to reach them will be considered non-compliant and will be withdrawn from the study, but their data up to the point of withdrawal will still contribute to the final analysis.

Data management

Data collection will begin on the day a participant is assigned to an intervention arm and continue until the trial termination or until participant withdrawal at any time for any reason. Data collection and continuous data entry will be prospectively performed by trained research assistants at baseline and during follow-up visits using an electronic case report form (eCRF) designed in the REDCap database to collect information on sociodemographic characteristics, sexual and reproductive histories, adherence and adverse events. Real-time data monitoring will be carried out by an experienced data manager, and queries will then be actively pursued to ensure prompt clarification. The research assistant will transfer the original data to the REDCap database. Access to the study dataset is restricted only to the PI. An independent steering committee will monitor and ensure strict compliance with the study protocol. Trial registration, progress or protocol modification will be documented in the Pan African Clinical Trial Registry (PACTR) and reported to LUTH’s ethics committee.

Statistical considerations

Data will be exported from the REDCap database to Stata version 18.0 for Windows (StataCorp LLC, Texas, USA) for statistical analyses. Missing or incomplete data will be handled by running a missing values analysis using pairwise or listwise deletion if no pattern is detected or the multiple imputation method if a pattern is detected [21]. The primary analysis will be conducted on an intention-to-treat and per-protocol basis. Descriptive statistics will be used to describe the characteristics of the participating women, and quantitative data will then be tested for normality with the Kolmogorov–Smirnov test with Lilliefors' significance correction. Comparison between the treatment arms at baseline will be performed using the independent samples t-test, Rank-sum test, Pearson's Chi-square or Fisher's exact test. To estimate the association between the treatment and the risk of primary (treatment response) and secondary outcomes (reversal of abnormal cervical cytology, HPV clearance, adverse events, and CIN2/3 recurrence), generalized linear models with a log link function and binomial distribution will be used to calculate risk ratios (RRs) and their 95% confidence intervals (CIs) while adjusting for all potential confounders. If convergence issues arise with the log-binomial model, we will apply a Poisson regression with robust standard errors to provide a valid approximation of the RR and CIs. The absolute risk reduction will be calculated as the percentage of women with treatment response in the LEEP arm subtracted from the percentage of patients with treatment response in the 5-FU/imiquimod arm. The number needed to treat (NNT) is then calculated as the inverse of absolute risk reduction. In predefined subgroup analyses, log-binomial models will be conducted stratified by age (< 50 vs ≥ 50 years), CIN grade (2 vs 3), and HIV status (positive vs negative) to evaluate the extent of variation in treatment effect on primary outcome based on these contextual factors, and *p*-values will be obtained from likelihood ratio tests.

Participants' recruitment and retention strategies

These include (1) a review of the study protocol by a participants panel and advocacy group to provide suggestions for ensuring that the study and recruitment approach is acceptable and feasible for patients; (2) developing clear, concise, and engaging messages that highlight the importance of the study and its potential benefits to participants and society; (3) utilizing community organizations, patient advocacy groups, and healthcare providers to reach potential participants; (4) using multiple recruitment sites if necessary to screen and enrol participants into the trial; (5) ensuring that study staff are trained in cultural competence and can communicate effectively with participants from different backgrounds; (6) ensuring participants and study staff feedback to improve recruitment efforts continuously; (7) printing and posting of sensitization posters and fliers within and around the recruitment clinics; (8) completion of the participant's locator form for each enrolled woman and review at each follow-up visit to ensure that the participant's contact information is up to date; (9) using of automated contact and scheduling of follow-up through short text message reminders; (10) assigning study staff to monitor scheduling and ensure participants tracking through personal phone calls 72 hours before or after (for failure to show) their scheduled follow-up visits; (11) reimbursement of transportation cost at 1,500 naira for each study visit; (12) ensuring treatment schedule is as convenient as possible and communicating the benefits and expectations of the topical treatment; (13) offering flexible scheduling, and transportation assistance, and providing thorough education about the importance of adherence to the treatment protocol; (14) evaluation and treatment of participants with CIN at no cost to them.

Plan to secure trial drugs and quality assurance

LEEP is the standard of care for treating CIN2/3 in the study site and will be available to trial participants at no cost. The study drugs (5-FU and Imiquimod cream) are currently available and approved by the National Agency for Food and Drug Administration and Control (NAFDAC) for clinicians and patients to use for various indications in Nigeria. Specifically, 5-FU cream is used for treating pre-cancerous and cancerous skin growths, while imiquimod cream is used for treating external genital and perianal warts. We have established a preliminary agreement with a reputable pharmaceutical company with extensive experience in providing clinical trial materials in Nigeria to supply the study drugs. We will follow a stringent procurement process, including verifying the credibility and compliance of suppliers with regulatory standards and ensuring proper documentation and certification for each batch of trial drugs. To maintain the highest quality standards, we will ensure that the trial drugs are (1) manufactured in a facility that complies with Good Manufacturing Practice guidelines and (2) stored under optimal conditions as specified by the manufacturers. Regular audits of storage facilities and periodic random sampling and testing of trial drugs will be conducted by the study Pharmacist throughout the study to ensure ongoing quality and consistency.

Plan to improve 5-FU/Imiquimod compliance

Strategies we plan to implement to improve treatment adherence in the study include (1) comprehensive education sessions to inform patients about the importance of adherence to the treatment protocol and the potential benefits of the experimental treatment; (2) use of automated reminders through phone calls, text messages, or emails to remind patients of their upcoming clinic visits and treatment schedules; (3) offering transportation support or reimbursement for travel expenses to reduce the burden of frequent clinic visits on patients; (4) regular monitoring and check-ins by research staff to address any concerns or side effects promptly and to reinforce the importance of adherence; (5) provision of small incentives, such as gift cards or other tokens of appreciation, to encourage compliance with the treatment protocol.

Ethical considerations

Approval for the trial (*TOPFIN-CIN Study*) was obtained from the research ethics committee of the Lagos University Teaching Hospital (ADM/DSCST/HREC/APP/7100 – April 25, 2025). The trial and statistical analysis plan will be conducted with fidelity. The purpose and nature of the study will be explained to all potential participants, and the willing participants will sign an informed consent form. The trial will adhere to the guidelines outlined in the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) checklist for reporting.

Quality control and data monitoring

All investigators and research assistants will be required to undergo training, including good clinical practice (GCP) training, before the trial to guarantee consistent practice. The training will cover a comprehensive understanding of the inclusion/exclusion criteria, follow-up protocols, and questionnaire completion. The research assistant will transfer identifiable data to an electronic database system housed in a secure facility at the trial site. Access to identifiable data will be limited solely to the principal investigator (KSO) throughout and after the trial concludes. The trial will undergo monitoring by quality assurance personnel from the research management office of the College of Medicine, University of Lagos, who will operate independently from the study team. Regular monitoring will occur to ensure the accuracy and quality of data throughout the study duration. Monitors will oversee and verify the essential documents (consent information, enrolment records, protocol deviations, and instances of loss to follow-up) for accuracy and completeness. An independent data and safety monitoring committee (DSMC) will also review the interim analyses to ensure the robustness of the data, treatment safety, and adherence to the trial protocol. Condensed study timelines are as illustrated in Table 2.

Dissemination of information

The trial will adhere to the reporting guidelines outlined in the Consolidated Standards of Reporting Trials (CONSORT) checklist, and the findings will be published in a peer-reviewed scientific journal. Any significant modifications to the trial protocol will be promptly communicated to the study investigators, LUTH-HREC, trial participants, funding agency, and the trial registry (PACTR).

Trial Status

This manuscript details protocol version 4.0 dated May 14th, 2025. Recruitment for the trial has not yet begun as of the point of manuscript submission. Enrolment of participants is scheduled to commence in July 2025, with the final participant expected to be included in the trial by June 2027. The anticipated completion date for the trial is September 2027.

Discussion

ICC remains the most prevalent HPV-associated malignancy and is a significant public health issue in Nigeria [2]. CIN2/3 are precancerous lesions of the cervix resulting from persistent HPV infection [2]. CIN2/3 lesions, if left untreated, can progress to ICC [3]. While excisional procedures such as LEEP and CKC are effective in treating CIN2/3 [3], they are associated with risks of complications including bleeding, infection, and cervical stenosis. They can lead to adverse obstetric outcomes, including cervical insufficiency and preterm births, in future pregnancies [4]. They are also associated with increased need for trained personnel, specialised equipment, and infrastructure, often limiting their accessibility, particularly in low-resource settings [5]. A non-surgical or medical treatment option with the potential for patient self-application could overcome many barriers to surgical therapy, including desire for childbearing, cost, health provider skill, equipment, location, and patient fear [6,7]. Additionally, medical treatment would address a treatment bottleneck in resource-limited settings by allowing immediate or point-of-care treatment of CIN2/3 [8]. This protocol, therefore, describes a non-inferiority, randomised, open-label, controlled trial that will be conducted at the gynecologic oncology, cytology and colposcopy clinics of LUTH in Lagos, Nigeria over two years (July 2025 to June 2027) to investigate the efficacy and safety of topical 5-FU/Imiquimod compared to LEEP (as standard of care) in treating patients with CIN 2/3. We have powered the study to detect the primary outcome to show that participants aged 30–65 years who had treatment with topical 5-FU/Imiquimod will achieve complete regression of CIN2/3 lesions at 3 months post-treatment initiation. If proven effective and safe, topical 5-FU and imiquimod could offer a non-invasive alternative to traditional excisional procedures, particularly in resource-limited settings where access to specialised surgical care is limited. By mitigating the risks associated with invasive procedures such as LEEP and CKC, this novel treatment approach could improve patient outcomes, preserve fertility, and alleviate the economic burden of ICC treatment on healthcare systems worldwide. Additionally, the study may contribute to the growing body of evidence supporting the use of immunomodulatory agents in treating HPV-related neoplasia, paving the way for future research and therapeutic advancements in the field. A limitation of this trial is that it is not statistically powered to detect the safety endpoint (although this is a planned secondary outcome). However, it will still generate essential safety data to inform future, larger phase 3 studies designed to validate the findings from this study more definitively.

Declarations

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Ethics approval and consent to participate

The authors declared no conflicts in the publication of this trial protocol.

Consent for publication

'Not applicable'.

Author contributions

KSO and OS made substantial contributions to the concept or design of the work in this paper. KSO and AIF drafted the article. All the authors (KSO, TVA, OO, AIF, IYA, PA, EOO, OAA, BA, OS, AAO, RIA, and JSB) revised the manuscript critically for important intellectual content and approved the version to be published.

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Competing interests

The author declares no competing interests in the publication of this article.

Availability of data and materials

No data are associated with this article. The authors intend to grant public access to the full protocol, participant-level dataset, and statistical code associated with this study.

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Tables

Table 1 and 2 are available in the Supplementary Files section.

Figures

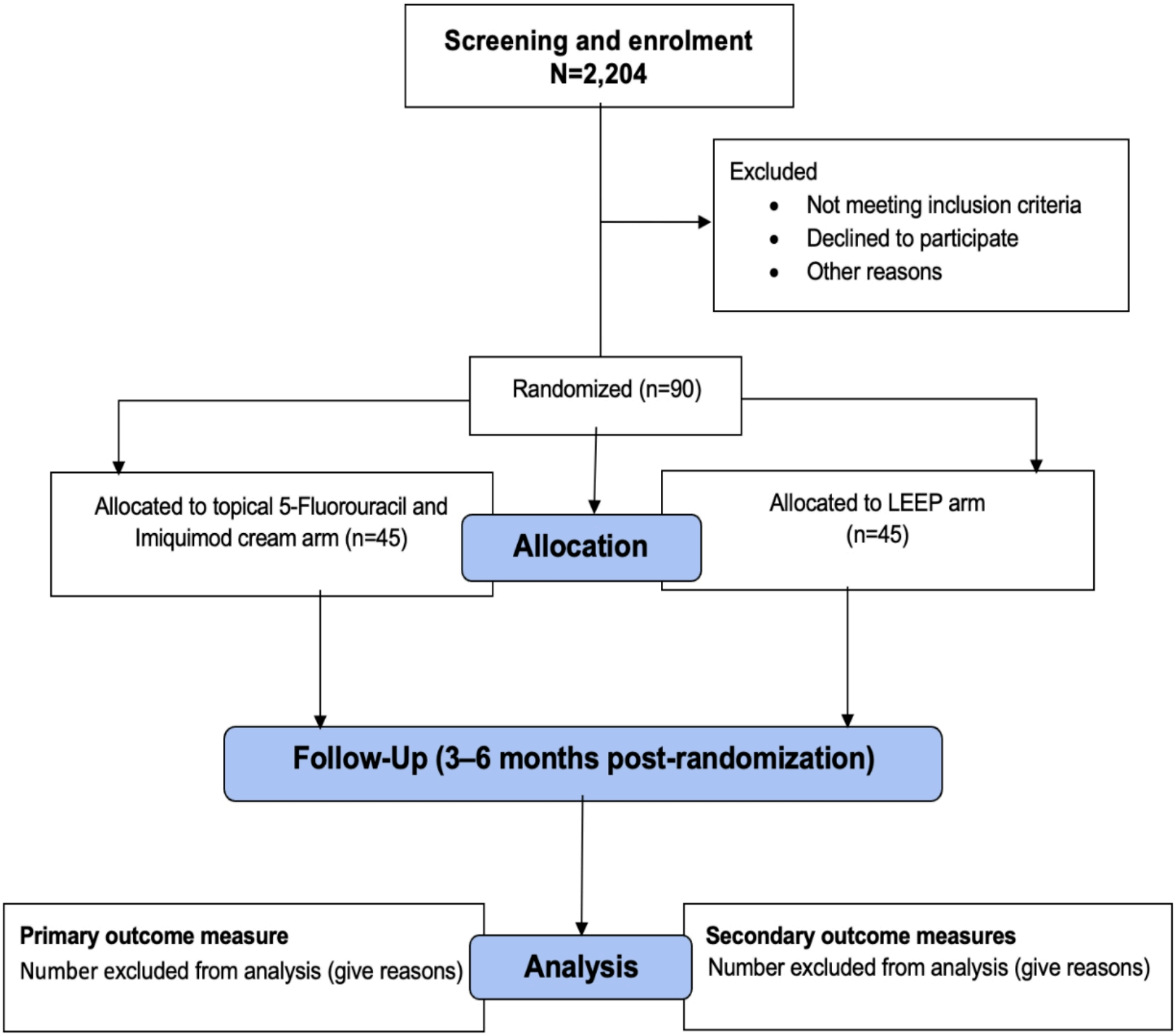

Figure 1: Trial flow chart

Figure 1

See image above for figure legend

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